Long-Term PPI Use: Weighing the Benefits and Risks

Proton pump inhibitors (PPIs) lower the amount of acid your stomach makes. They work very well, and many people truly need them for months or years. Some studies link long-term use to health problems, but the strongest type of study (randomized controlled trials) has not shown more of these problems in people taking PPIs. The links come from observational studies, which cannot prove the medicine caused the problem.

Important

Do not stop your PPI on your own. Talk with your provider about whether you still need it, and at what dose.

What PPIs are

Common PPIs include omeprazole (Prilosec), esomeprazole (Nexium), pantoprazole (Protonix), lansoprazole (Prevacid), rabeprazole (Aciphex) and dexlansoprazole (Dexilant). Some are sold over the counter. They are used for heartburn and reflux (GERD), stomach and intestinal ulcers, and to protect the stomach in people at high risk of bleeding. The most common side effects, even in the short term, are headache, diarrhea and stomach pain.

The benefits: who may need a PPI long term

Long-term use is usually appropriate if you have one of these conditions:

- Severe damage to the esophagus (food pipe) from acid, such as severe erosive esophagitis, an esophageal ulcer, or a narrowed esophagus (stricture)
- Barrett's esophagus
- Eosinophilic esophagitis (in some people)
- A high risk of bleeding in the stomach or intestines, for example a past bleeding ulcer, or taking aspirin, blood thinners or anti-inflammatory pain medicines (NSAIDs) along with other risk factors

- Heartburn that comes back every time you try to stop, confirmed by your provider

The benefit
For these conditions, a PPI heals damage, prevents bleeding and complications, and relieves symptoms. In most of these cases the benefits outweigh the possible risks.

The risks: what studies have found

Observational studies have linked long-term PPI use to the problems below. Observational studies only show that two things happen together, not that one causes the other. People who take PPIs for years often have other health conditions and risk factors that raise the chance of these same problems, so the PPI may not be the cause. Randomized controlled trials have not shown a higher rate of any of these problems in PPI users.

Concern	What studies suggest	What this means for you
Chronic kidney disease	Linked in observational studies; cause not proven	Tell your provider if you have kidney disease. Report less urine, swelling, fever or rash
Broken bones (especially hip)	Link is stronger with higher (double-strength or twice-daily) doses; cause not proven	Use the lowest dose that works
Dementia and memory decline	Linked in observational studies; cause not proven	Review whether you still need the PPI
Pneumonia and C. difficile (a serious bowel infection with watery diarrhea)	Link is stronger with higher doses; cause not proven	Report diarrhea lasting more than a few days, especially after antibiotics

Concern	What studies suggest	What this means for you
Low B12, magnesium, calcium or iron	Linked in studies	Routine blood tests and extra supplements are not recommended for most people. Report numbness, tingling, muscle cramps, a fluttering heartbeat or unusual tiredness
Heart problems	Linked in observational studies; cause not proven. Omeprazole and esomeprazole may weaken the blood thinner clopidogrel (Plavix)	If you take clopidogrel, ask your provider which PPI is best for you
Stomach polyps, and stomach, colon or pancreas cancer	Linked in observational studies; cause not proven	Talk with your provider if this worries you

The bottom line
These concerns are worth reviewing, but they are unproven. For people with a clear reason to take a PPI, the risk of stopping is greater than the possible risks of continuing.

The risks of taking a PPI you don't need, or stopping one you do

If you...	What can happen
Take a PPI with no ongoing reason	More pills, extra cost, possible side effects, and rebound heartburn when you stop. A trial of reducing or stopping is reasonable.
Have a reason to take it (such as Barrett's esophagus, severe esophagitis or a high bleeding risk) and stop	Symptoms can return and serious problems can follow, including life-threatening bleeding.

If you...	What can happen
Stop only because you fear side effects	Fear of side effects alone is not a reason to stop. Nearly 4 in 10 patients try to stop because of safety worries, and most do so without talking to a provider.
Take a higher dose (twice a day or double strength)	Studies link higher doses to more concerns. Ask whether you can step down to once a day.

Do I still need my PPI?

Experts recommend that your provider review your PPI regularly to confirm why you take it. Ask about:

- **Stepping down:** if you take it twice a day, many people can drop to once a day
- **Lowering the dose** or using it only when symptoms flare
- **A trial off the medicine,** if you have no clear long-term reason to take it
- **Other options,** such as an H2 blocker like famotidine (Pepcid), for milder symptoms

If you do stop after long-term use, heartburn or stomach upset may come back for a few weeks. This is called rebound acid, and it is temporary. It does not always mean you need the PPI back. Your provider can give you a plan to get through it.

Ways to lower your risk

- Take your PPI 30 to 60 minutes before a meal, as directed
- Use the lowest dose that controls your symptoms, and ask about once-a-day dosing if you take it twice a day
- Have your PPI reviewed regularly by your primary care provider
- Avoid NSAIDs (ibuprofen, naproxen) when you can; ask before using them with a PPI
- Reflux often improves with weight loss, smaller meals, not eating within 3 hours of bedtime, raising the head of your bed, and limiting alcohol, tobacco and personal trigger foods
- Eat a balanced diet and stay active. You do not need extra calcium, B12 or magnesium supplements, bone density scans, routine blood tests, or probiotics just because you take a PPI. Tell your provider if you have any of the symptoms listed below

When to call your provider

How urgent	Signs to watch for
Get care right away	Black, tarry or bloody stools; vomiting blood; chest pain or pressure (**call 911**, and do not assume it is heartburn)
Call your provider soon	Trouble or pain with swallowing, or food getting stuck; unplanned weight loss or repeated vomiting; severe or lasting watery diarrhea; muscle cramps, weakness or a fluttering heartbeat; numbness, tingling or unusual tiredness; passing much less urine, new swelling, fever or rash

My PPI plan

Item	Details
My PPI and dose	
Why I take it	
Plan after today's visit	Continue as is / Lower the dose / Trial off
Labs or follow-up	
Next review date	

This handout is general education. It does not replace advice from your own provider.

Sources

- AGA Clinical Practice Update on de-prescribing of proton pump inhibitors: expert review. Gastroenterology, 2022
- Long-Term Use of Proton-Pump Inhibitors: Unravelling the Safety Puzzle. Cureus, 2024
- Proton Pump Inhibitors and Cancer Risk: A Comprehensive Review of Epidemiological and Mechanistic Evidence. Journal of Clinical Medicine, 2024

- Long-term use of proton pump inhibitors as a risk factor for various adverse manifestations. Therapie, 2021
- The risks of long-term use of proton pump inhibitors: a critical review. Therapeutic Advances in Drug Safety, 2019
- The Risks and Benefits of Long-term Use of Proton Pump Inhibitors: Expert Review and Best Practice Advice From the American Gastroenterological Association. Gastroenterology, 2017

Based on the references cited in the Dr. Oracle article "What are the concerns associated with long-term proton-pump inhibitor (PPI) use?"